

Software Requirements Specification (SRS) & User Manual

Part 1: Software Requirements Specification (SRS)

1. Introduction

This document outlines the Software Requirements Specification (SRS) and the User Manual for the DTU Student Data Pipeline. The application is designed to ingest raw student datasets, automatically clean them, and provide an interactive dashboard for filtering and shortlisting candidates.

2. Functional Requirements

FR1. Data Ingestion: The system shall allow users to upload CSV files containing student records.

FR2. Auto-Cleaning: The system shall clean quotes from names, remove 'Grade' text from grade values, impute missing marks with 0, clip out-of-range marks (0-100), remove missing names, remove exact duplicates, normalize gender formats, and strictly recalculate the Total column.

FR3. Debar Management: The system shall provide a real-time checkbox toggle to mark a student as Debarred, which immediately excludes them from analytics.

FR4. Shortlist Filtering: The system shall allow filtering by minimum Total, Math, Science, and English scores.

FR5. Data Export: The system shall allow downloading the final shortlist as a CSV file.

3. Non-Functional Requirements

NFR1. Usability: The interface shall use a modern, glassmorphic UI design and be accessible via a standard web browser.

NFR2. Performance: File processing and data cleaning shall occur in real-time upon upload.

Delhi Technological University - Student Data Pipeline

Part 2: User Manual

1. Getting Started

Ensure you have Python 3.8+ installed. Run the application locally by executing ``streamlit run app.py`` in your terminal. The application will open automatically in your browser at <http://localhost:8501>.

2. Data Upload & Cleaning

- Navigate to the 'Data Upload & Cleaning' tab.
- Drag and drop your raw CSV file into the upload zone.
- The system will instantly display a 'Data Cleaning Report' expanding on how many duplicates, ghost records, and missing values were fixed.

3. Debarring Candidates

- In the same tab, you will see the Candidate Roster.
- Check the 'Debarred?' box next to any student to instantly remove them from consideration. This state is preserved across tabs.

4. Analytics & Shortlisting

- Switch to the 'Analytics & Shortlisting' tab.
- Use the top 'Shortlist Filters' to set minimum score thresholds.
- View the interactive scatter plots, histograms, and box plots to analyze the shortlisted cohort.
- Click 'Export Final Shortlist as CSV' at the bottom to download your results.